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State Anxiety Responses to Acute Exercise in Women With High Social Physique Anxiety

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This study examined the state anxiety (SA) and perceived arousal (AS) responses to self-selected or imposed-intensity bouts of acute exercise performed in different environments by 30 women with high social physique anxiety (SPA). Participants were randomly assigned to a self-selected or imposed-intensity choice group and subsequently (a) exercised in a naturalistic environment, (b) exercised in a laboratory environment, and (c) rested quietly. Assessments of SA and AS were obtained before, during, and following each condition and data were analyzed via separate $2 \times 3 \times 6$ (Intensity Choice $\times$ Condition $\times$ Time) ANOVAs with repeated measures on the Condition and Time factors. Results revealed that AS increased during both exercise conditions. Conversely, SA was elevated only during the naturalistic exercise condition, and item-level analysis revealed that this increase was composed of both activation and apprehension/worry inventory items. Although SA was reduced 5 min following both exercise conditions, the anxiolytic effect only persisted following the laboratory condition. No significant differences in SA or AS were observed between the two groups. It is concluded that environmentally-induced perceived evaluative threat is associated with an increase in SA during exercise and may attenuate postexercise anxiolysis in women with high SPA.

Key Words: anxiolysis, physical activity, self-presentational concerns

In today's society, the attempt to make a positive impression on others is particularly salient in regard to physical attractiveness (Leary, Tchividjian, & Kraxberger, 1994). Consequently, many individuals exercise for self-presentational purposes such as weight control, enhanced muscle tone, and improved overall appearance (Conroy, Motl, & Hall, 2000; Silberstein, Striegel-Moore, Timko, & Rodin, 1988). Although the desire to improve one's appearance may motivate some individuals to exercise, self-presentational concerns about physique may also deter participation among individuals who feel they are not physically attractive to others. For example, it has been suggested that social physique anxiety (SPA), the

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apprehension one feels in response to others' evaluation of his or her appearance, impedes exercise participation (Hart, Leary, & Rejeski, 1989). Consistent with this contention, an inverse relationship between SPA and exercise adherence has been documented (Lantz, Hardy, & Ainsworth, 1997; Treasure, Lox, & Lawton, 1998). Furthermore, obese women cite their apprehension about being evaluated by others as the primary reason for avoiding public exercise settings (Bain, Wilson, & Chaikind, 1989), and individuals with heightened SPA prefer to exercise in private settings or environments where less revealing attire is worn (Crawford & Eklund, 1994; Spink, 1992).

SPA demonstrates a reliable association with the antecedents of exercise behavior such as motives, preferences, and attitudes toward physical activity (Conroy et al., 2000; Crawford & Eklund, 1994; Hausenblas & Martin, 2000). Nevertheless, we know little about the psychological consequences of exercise participation among individuals with SPA. For example, whereas exercise-induced affective responses represent a fundamental domain through which self-presentational processes may impact exercise behavior (Leary, 1992), for individuals with SPA the affective responses to exercise have yet to be systematically investigated. Hence the primary objective of this study was to examine state anxiety responses to self-selected and imposed-intensity bouts of acute exercise performed in environments characterized by differing levels of evaluative threat among women with high SPA.

State Anxiety Responses to Acute Exercise

The Role of SPA and Perceived Evaluative Threat

Findings from the extant literature suggest that acute bouts of aerobic exercise are associated with improvements in a variety of affective outcomes, the most commonly studied of which has been reductions in state anxiety (Breus & O'Connor, 1998; Petruzzello, Jones, & Tate, 1997; Raglin & Wilson, 1996; Youngstedt, O'Connor, Crabbe, & Dishman, 1998). However, it is important to recognize that virtually all of the existing research has been conducted in samples characterized by normative or low levels of anxiety. In light of this fact, many researchers have recently questioned the validity of studying the anxiolytic responses to exercise in the absence of naturally occurring or experimentally-induced anxiety (Ekkekakis, Hall, & Petruzzello, 1999; Focht, 2002; O'Connor, Raglin, & Martinsen, 2000).

In accordance with these concerns, there have been attempts in recent investigations to heighten anxiety prior to exposure to the exercise stimulus (Acevedo, Dzewaltowski, Kubitz, & Kraemer, 1999; Bartholomew, 1999; Youngstedt et al., 1998). It is important to recognize that it remains unclear as to whether any of the previously employed anxiogenic procedures resulted in an appraisal of threat. Consequently, the validity of these methods for inducing state anxiety has yet to be established and the efficacy of alternative anxiogenic procedures warrants further study.

In this regard, research on self-presentational theory has revealed that exposure to socially evaluative settings results in heightened anxiety and perceived stress among individuals with self-presentational concerns (Hart et al., 1989; Schlenker & Leary, 1982). Given the prospect for negative social evaluation that exists in most public fitness facilities (Leary, 1992), examining state anxiety re-

sponses to exercise undertaken in a potentially threatening environment—among individuals with self-presentational concerns—would provide a theoretically relevant paradigm for investigating exercise-induced anxiety.

Consistent with the conceptual underpinnings of self-presentational theory (Leary, 1992), state anxiety responses to acute exercise would be expected to vary with the evaluative threat present in the environment as perceived by persons high in SPA. In support of this theoretical position, women with high SPA reported lower self-efficacy, a cognitive determinant of exercise-induced affective responses, during stationary cycling undertaken in a self-presentational exercise environment (Katula, McAuley, Mihalko, & Bane, 1998).

Alternatively, Focht and Hausenblas (2001) observed that state anxiety responses to acute exercise undertaken in a naturalistic environment did not differ as a function of SPA. However, because in-task anxiety responses were not obtained, it is unclear whether the naturalistic exercise environment fostered perceptions of evaluative threat. Thus, while this finding suggests that the anxiolytic effect of acute exercise is not moderated by SPA, it is possible that individuals with high SPA may not experience an exercise-induced reduction in state anxiety in an environment that has a prospect of evaluative threat (Leary, 1992). Nevertheless, this theoretical prediction has yet to be empirically tested and the anxiolytic benefits of acute exercise among individuals with high SPA warrants further exploration.

The Role of Self-Selected vs. Imposed Exercise Intensity

In the majority of studies examining the affective responses to acute exercise, intensity during the bout has been imposed upon the exercisers. Accordingly, relatively little is known about the affective responses to acute exercise performed at a self-selected exercise intensity. Given that individuals choose the intensity of activity during habitual exercise participation, the ecological validity of psychological responses to imposed-intensity exercise remains unknown. In accordance with such concerns, identifying the consequences of imposed vs. self-selected intensity exercise has been advocated as an important consideration in developing conceptual models of the exercise/affect relationship (Ekkekakis & Petruzzello, 1999; Morgan, 1997).

Findings from prior research specifically addressing the anxiolytic effects of exercise have shown that state anxiety is reduced following acute aerobic exercise undertaken at a self-selected intensity (Dishman, Farquhar, & Cureton, 1994; Farrell, Gates, Maksud, & Morgan, 1982; Raglin & Morgan, 1987). However, each of these studies involved exercise undertaken in the absence of naturally occurring or experimentally-induced anxiety. Thus, direct comparisons of the effects of self-selected and imposed-intensity exercise upon state anxiety have yet to be systematically investigated and further research on this issue has been advocated (Ekkekakis & Petruzzello, 1999; Morgan, 1997; Rejeski, 1994).

Use of the State Anxiety Inventory

The State Anxiety Inventory (SAI) of the State-Trait Anxiety Inventory (Spielberger, Gorsuch, Lushene, Vagg, & Jacobs, 1983) is the most frequently used measure for examining the anxiolytic effects of acute exercise (O'Connor et al., 2000). However, despite extensive use of the SAI as reported in the exercise

psychology literature, concerns have emerged regarding the psychometric integrity of the inventory when administered in conjunction with acute exercise. For example, analysis of change in individual SAI items during acute exercise revealed that the increase in the scores of items related to perceived activation was accompanied by a decrease in the scores of items reflecting the cognitive elements of anxiety (Rejeski, Hardy, & Shaw, 1991).

Response divergence between SAI items representing cognitive appraisal and perceived activation has also been observed in more recent studies (Ekkekakis et al., 1999; Katula, Blissmer, & McAuley, 1999; McAuley, Mihalko, & Bane, 1996). Indeed, a principal-components analysis of postexercise SAI responses conducted by Ekkekakis et al. (1999) revealed a multicomponent structure that differentiated items reflecting activation from items tapping apprehension or worry. Ekkekakis et al. concluded that the divergent pattern of change in the activation and apprehension/worry items confounds the total SAI score during participation in vigorous exercise. Based on these findings, it is clear that further inquiry on the use of the SAI during acute exercise is warranted. Contrasting SAI item responses with changes in an index of perceived activation during exercise conditions that impose different levels of evaluative threat and physical demand would provide additional information as to the validity of the SAI in the context of acute exercise.

The Present Study

In summary, given the salience of self-presentational processes in exercise behavior (Conroy et al., 2000; Leary, 1992), examining changes in state anxiety during and following acute exercise among women with high SPA would help delineate the affective responses to physical activity. More specifically, since many naturalistic exercise settings have qualities that may foster self-presentational anxiety, it is possible that the anxiolytic effect commonly observed following acute exercise might not emerge in women with high SPA when exercising in environments that present the prospect of evaluative threat.

For women with high SPA, it should be acknowledged that while perceptions of evaluative threat may have an impact on state anxiety during exercise, the influence of such social environmental factors may be related to the intensity of the exercise stimulus. Indeed, the social environmental context within which exercise is undertaken has been shown to influence the selection of exercise intensity and perceptions of physical exertion (Rejeski, 1985). Additionally, Worringham and Messick (1983) reported that individuals tended to run at a faster speed when the prospect of social evaluation was present, and self-presentational concerns have also been found to influence ratings of perceived exertion during exercise (Boutcher, Fleischer-Curtain, & Gines, 1988).

These findings suggest that socially evaluative qualities of the environment in which exercise is performed influence perceptions of intensity and exertion during exercise. Yet no prior research has directly addressed the implications of this issue for the anxiolytic effect of acute exercise. The only extant research examining the state anxiety responses to acute exercise among women with high SPA employed self-selected intensity exercise (Focht & Hausenblas, 2001). Consequently, the role played by choice of exercise intensity in this relationship remains unclear and research is warranted on the influence of environmentally-induced perceptions of evaluative threat and the choice of exercise intensity (i.e., imposed

vs. self-selected intensity) upon the state anxiety responses to acute exercise. The purpose of this investigation was to examine the influence—upon state anxiety responses in low-active women with high SPA—of self-selected and imposed-intensity bouts of acute aerobic exercise undertaken in different environments. It is hypothesized that, due to the prospect of evaluative threat, the naturalistic exercise condition will be associated with an increase in state anxiety and that anxiolytic benefits will only emerge following exercise undertaken in the laboratory environment. Additionally, because low-active participants may perceive imposed-intensity exercise as aversive, it is hypothesized that self-selected intensity exercise will result in more favorable state anxiety responses.

Method

Participants

Thirty female college students (77% Caucasian, mean age 20.20 yrs $\pm$ 1.65) were recruited from lab and lecture courses at a university in the southeastern U.S. to participate in this study. In previous research, reductions in state anxiety following acute exercise in women with high SPA were associated with moderate ($-.51$) to large ($-.86$) effect sizes (Focht & Hausenblas, 2001). The results of this prior research were used as the basis for an a-priori sample size estimation. Using an alpha level of .05, a moderate effect size of .50, and a power of .80, this estimation procedure indicated that the sample size employed provided adequate power for detecting the main effects and interactions of interest (Kraemer & Thieman, 1987; Tran, 1997). All volunteers received extra course credit for their participation. An exclusively female sample was used based on findings indicating that women report higher SPA (Hart et al., 1989). To obtain a sample of high-SPA women with low physical activity levels, we assessed self-reported levels of SPA and physical activity participation prior to including them in this study. The inclusion criteria were based on the findings of prior research suggesting that less active women characterized by self-presentational anxiety would be the most responsive to manipulations of evaluative threat (Hart et al., 1989; Leary, 1992). Selected demographic characteristics are listed in Table 1.

Volunteers scoring 36 or higher on the Social Physique Anxiety Scale (Hart et al., 1989) who also reported undertaking two or fewer bouts of moderate or vigorous exercise during a typical week were eligible to participate. The high SPA value for eligibility is approximately one standard deviation above the average SPAS scores of women reported in previous research (Hausenblas & Martin, 2000; Martin, Rejeski, Leary, McAuley, & Bane, 1997). The low-active criterion was determined by self-report of exercise levels that fall below the threshold for achieving health and fitness benefits (U.S. Dept. Health & Human Services, 1999). All volunteers completed a medical history questionnaire and an informed consent document, and received extra class credit for their participation.

Measures

Social Physique Anxiety Scale. The Social Physique Anxiety Scale (SPAS) was used to assess the anxiety of participants as a function of others' evaluation of their physique (Hart et al., 1989). However, since recent research indicates that a revised version of the original 12-item inventory demonstrates the most sound

Table 1 Demographic Characteristics, Perceived Evaluative Threat, RPE, and Exercise Intensity of Exercise Groups

Variable	Self-Selected (<i>n</i> = 15)		Prescribed-Intensity (<i>n</i> = 15)		ES
	<i>M</i>	(<i>SD</i>)	<i>M</i>	(<i>SD</i>)	
Age (yrs)	20.0	(1.4)	20.4	(1.8)	
Height (cm)	157.9	(6.33)	159.9	(4.69)	
Weight (kg)	70.0	(9.34)	69.1	(8.90)	
Exercise sessions per week	1.2	(.86)	1.4	(1.0)	
Perceived Evaluative Threat					
Naturalistic	2.4	(.92)	2.7	(.83)	
Laboratory	1.4	(.51)	1.8	(.70)	
Quiet rest	1.0	(.38)	.93	(.33)	
Average Exercise Intensity (% of HR max)	63.5	(5.7)	78.3	(2.2)	3.43
Perceived Exertion					
5 minutes	12.1	(2.0)	13.6	(1.6)	.83
10 minutes	12.8	(2.0)	14.1	(1.8)	.68
15 minutes	13.3	(2.0)	14.5	(1.7)	.65
20 minutes	13.6	(2.0)	14.7	(1.7)	.59

Note: ES = effect sizes

psychometric properties, the abbreviated 9-item version of the SPAS was used in this investigation (Martin et al., 1997). In completing the scale, participants indicated on a 5-point Likert scale the degree to which each item was characteristic of them. The SPAS scores range from 9 (low) to 45 (high). Research has found that the 9-item SPAS demonstrates acceptable convergent and divergent validity and adequate internal consistency (Hausenblas & Martin, 2000; Martin et al., 1997). Consistent with these findings, the SPAS exhibited good internal consistency in this study ($\alpha = .86$). Whereas we employed the 9-item SPAS (Martin et al., 1997) to quantify SPA, it should be noted that a consensus as to the most appropriate version of the scale has yet to be established (Motl & Conroy, 2000). Although a comprehensive evaluation of the psychometric properties of the SPAS is outside the scope of this study, further exploration of the validity and factorial invariance of this inventory is warranted.

Leisure-Time Exercise Questionnaire. The Leisure-Time Exercise Questionnaire (LTEQ) measures the self-reported frequency of strenuous, moderate, and mild bouts of exercise during a typical week (Godin & Shephard, 1985). LTEQ subscale scores are converted into metabolic equivalents [METs; (Mild $\times$ 3) + (Moderate $\times$ 5) + (Strenuous $\times$ 9)] and summed to provide an estimate of total

MET expenditure from exercise for an average week. Previous research has shown that the LTEQ possesses adequate reliability and validity (Jacobs, Ainsworth, Hartman, & Leon, 1993).

State-Trait Anxiety Inventory. State anxiety was measured using the 20-item Form Y-1 (SAI) of the State-Trait Anxiety Inventory (STAI; Spielberger et al., 1983). The SAI has undergone rigorous validation procedures in nonexercise conditions (Spielberger et al.), and it has been used extensively in investigations of the state anxiety responses to acute exercise (O'Connor et al., 2000). The SAI demonstrated good internal consistency (alpha ranging from .83 to .91 across the assessments) within this sample. Participants responded to each item on a 4-point Likert scale ranging from 1 (not at all) to 4 (very much so), with higher scores indicating higher state anxiety.

Arousal Scale. Perceived arousal was assessed using the arousal scale (AS) of the affect grid (Russell, Weiss, & Mendelsohn, 1989). In completing the assessment, participants used the instructions outlined by Russell et al. and placed a single mark on the 9-by-9 grid which yields scores for both arousal and valence. Given the purpose of this investigation, we have limited our analysis to the AS. Scores on the AS range from one (low arousal) to 9 (high arousal). The AS has adequate validity and reliability (Russell et al., 1989) and has been shown to correlate in the expected direction with indices of perceived arousal with exercise research (VanLanduyt, Ekkekakis, Hall, & Petruzzello, 2000).

Ratings of Perceived Exertion. To quantify the participants' overall sense of effort during the exercise session, ratings of perceived exertion (RPE) were obtained using Borg's 6–20 scale (Borg, 1998). RPE has been conceptualized as a psychobiological configuration of sensations associated with the perception of effort during exercise, and it has been used extensively in previous research.

Ratings of Perceived Evaluative Threat. Participants were asked to rate their perception of evaluative threat in each environment. Consistent with procedures used in previous research (Hart et al., 1989), participants were asked to rate how threatening they perceived each session to be from the perspective of physique evaluation using a 5-point Likert scale ranging from 0 (not at all) to 4 (extremely).

Procedures

Thirty participants who met the eligibility criteria were randomly assigned to perform either self-selected intensity exercise ($n = 15$) or imposed-intensity exercise ($n = 15$). They completed three experimental conditions: (a) a bout of cycling in a naturalistic environment, (b) a bout of cycling in a laboratory, (c) and a quiet rest control condition. Each session lasted 20 min. The order in which the three conditions were presented was randomized and counterbalanced and the conditions were completed on separate days. To standardize the attire during each condition, participants had to wear a T-shirt and pair of shorts to each session. Those assigned to the imposed-intensity group completed each bout of cycling in both the laboratory and naturalistic conditions at an intensity of 70–80% of age-predicted maximum heart rate. Those assigned to the self-selected intensity group completed the bouts of cycling in each environment using a self-determined level of intensity (see Table 1).

The exercise protocol began with a 5-min warm-up for each group. Following the warm-up, participants in the self-selected intensity group determined the resistance and pedaling cadence throughout the exercise session. By contrast, those in the imposed-intensity group cycled at a cadence of 50 rpm as the pedal resistance was gradually increased. Upon achieving the prescribed heart rate range, participants then exercised at this intensity for the duration of the session. Pedal cadence and resistance were adjusted accordingly throughout the exercise session to maintain the target heart rate range.

Exercise in the naturalistic condition was performed at a university fitness facility adjacent to the exercise psychology lab. In contrast to the study by Focht and Hausenblas (2001), aspects of the naturalistic environment in the present study were manipulated in order to maximize the perception of evaluative threat. Specifically, consistent with procedures used previously (Katula et al., 1998), participants exercised on a stationary cycle in front of a full length mirror which allowed them to view themselves. An average of 36 ($SD = 12.27$; range = 18–62) other recreational exercisers (14 men, $SD = 5.69$; 22 women, $SD = 7.89$) were present in the fitness center during the naturalistic condition. Taken collectively, the constant presence of other exercisers, staff members, and the full length mirror created an environment that provided ample opportunity for participants to experience self-presentational concerns about their physique.

Exercise in the laboratory condition was undertaken in a private laboratory setting without mirrors or the presence of other exercisers. To maintain the private nature of the environment, interaction between the participant and the researcher was limited to the assessment procedures. At all other times during the laboratory condition the experimenter stayed out of view and refrained from interacting with the participant.

During the quiet rest condition, participants sat quietly in a room free of distractions within the exercise psychology laboratory. They were not given any specific instructions beyond being asked not to sleep during the session.

Assessments

Upon arriving at the laboratory, participants completed the baseline SAI and AS assessments and were fitted with the Polar Vantage heart rate monitor. In order to minimize the influence of demand characteristics or expectancy effects on the findings, participants were informed at the beginning of the study that it would be possible to receive any combination of exercise and quiet rest conditions across the three sessions. Furthermore, only after completing the baseline assessment were they told which condition they would be undertaking.

RPE and heart rate were recorded every 5 min during both exercise conditions and the AS was assessed prior to, every 5 min during, and 5 min following each condition. Assessments of the SAI were obtained prior to, during (15th minute of activity), and at 5, 60, 120, and 180 min following each condition. At the conclusion of each session the participants sat in a room free of distractions for 5 min while the first postcondition assessments of SAI and AS were completed. They then completed the 60-, 120-, and 180-min postcondition assessments of the SAI using a signal-contingent technique. Specifically, following the initial postcondition assessment, they were given a pager and three copies of the SAI and allowed to leave the lab. Participants were paged at each remaining assessment interval and

instructed to complete the SAI. To assess whether the performance of nonhabitual activity acted as a source of response bias during the signal-contingent assessments, participants also completed a behavioral questionnaire during each sampling. They recorded the activity they were performing at the time of the sampling and indicated whether it was part of their normal daily routine.

Although the signal-contingent assessments of the SAI were completed at three predetermined intervals, participants were not aware of when the pager tones would be sent. Rather, they were informed that the pager would be activated at random intervals three times during the 4 hours immediately following the completion of each condition. The signal-contingent technique allows for the assessment of psychological states within one's naturalistic environment. Therefore this technique provides additional information on the length of time that postexercise anxiety persists following the resumption of normal daily activities, resulting in a more ecologically valid assessment of state anxiety compared to conventional laboratory measurements (Gauvin, Rejeski, & Norris, 1996; Gauvin, Rejeski, & Reboussin, 2000).

Statistical Analysis

Ratings of perceived evaluative threat in each condition were analyzed with a one-way analysis of variance (ANOVA). To examine differences in exercise intensity and the perception of effort, analysis of the average exercise intensity during each session was conducted with a 2 (Intensity Choice: self-selected and imposed) $\times$ 2 (Condition: naturalistic and laboratory) ANOVA, and RPE data was analyzed using a 2 (Intensity Choice: self-selected and imposed) $\times$ 2 (Condition: naturalistic and laboratory) $\times$ 4 (Time: 5, 10, 15, 20 min) ANOVA. The SAI was analyzed using a 2 (Intensity Choice: self-selected and imposed) $\times$ 3 (Condition: naturalistic, laboratory, quiet rest) $\times$ 6 (Time: pre, during, post 5, 60, 120, 180) mixed-model ANOVA. The AS was analyzed using a 2 (Intensity Choice: self-selected and imposed) $\times$ 3 (Condition: naturalistic, laboratory, and quiet rest) $\times$ 6 (Time: pre, 5, 10, 15, 20 min during activity, and 5 min post) ANOVA.

The traditional *F* tests were computed using the Huynhdt and Feldt (1976) conservative degrees of freedom adjustments. Bonferroni adjustments yielded an alpha level of $p < .0125$ for all analyses conducted. Tukey's post-hoc tests were used to determine the location of mean differences when significant effects were detected, and effect sizes (ES) accompanying the changes in the SAI and AS were calculated by subtracting the postcondition means from the baseline values and dividing by the pooled standard deviation (Thomas, Salazar, & Landers, 1991).

Results

Perceived Evaluative Threat, Exercise Intensity, and Signal-Contingent Assessments

Analysis of ratings of perceived evaluative threat revealed a significant difference between conditions, $F(2, 86) = 35.28, p < .001$. It was found that participants rated the naturalistic environment significantly more threatening than the laboratory environment. Additionally, both exercise conditions were rated as more threatening than the quiet rest condition. Thus the manipulation was successful in

producing exercise environments that differed in the ratings of perceived evaluative threat (see Table 1).

Analysis of the average exercise intensity yielded a significant main effect for Intensity Choice, $F(1, 28) = 35.28, p < .001$. Additionally, the Intensity Choice $\times$ Condition interaction approached significance, $F(1, 28) = 5.66, p = .024$. The results revealed that exercise intensity was significantly higher during imposed-intensity activity (see Table 1). Although the interaction effect was not statistically significant, it should be noted that, for the self-selected intensity group, there was a trend toward higher intensity during exercise in the naturalistic environment.

Regarding the signal-contingent assessments, 227 of the 270 total samplings (84%) were completed while participants were engaged in their everyday activities. The assessments completed during activities not typically performed on a daily basis appeared in random fashion across participants and samplings, suggesting that postexercise responses were not influenced by the performance of nonhabitual activity. Consequently, the signal-contingent procedure was successful in assessing postexercise state anxiety responses following the resumption of normal daily activities. Furthermore, the results for the ratings of perceived evaluative threat, exercise intensity, and signal-contingent assessments indicate that the experimental procedures were successful for the purposes of this study.

Ratings of Perceived Exertion

Analysis of the RPE responses yielded a significant Intensity Choice $\times$ Condition interaction, $F(1, 28) = 6.45, p < .01$; as well as significant main effects for Time, $F(3, 84) = 33.12, p < .001$; Intensity Choice, $F(1, 28) = 38.05, p < .01$; and Condition, $F(1, 28) = 8.31, p < .007$. Results revealed that, in comparison to self-selected intensity exercise, RPE was significantly higher ($ES = .71$) during imposed-intensity exercise. For the imposed-intensity group, RPE during exercise did not differ between environments ($ES = .07$). However, for the self-selected intensity group, RPE was found to be significantly higher ($ES = .67$) during exercise undertaken in the naturalistic environment (see Figure 1).

State Anxiety

Analysis of the SAI yielded a significant main effect for Time, $F(5, 140) = 4.96, p < .001$, as well as a significant Condition $\times$ Time interaction, $F(10, 280) = 4.37, p < .001$. Post-hoc analyses revealed a significant increase from preexercise to during exercise in the naturalistic environment only. Significant reductions in state anxiety were observed from preexercise to 5 min postexercise in both environments. However, whereas the anxiolytic response persisted through the 120-min postexercise assessment following the laboratory environment, state anxiety did not differ significantly from baseline at any of the remaining assessments following exercise in the naturalistic environment. No significant changes in the SAI were detected following quiet rest (see Table 2). In summary, state anxiety increased during exercise in the naturalistic environment, and the anxiolytic effect observed 5 min after both exercise conditions only persisted following exercise in the laboratory environment.

To examine whether the total SAI scores during exercise were confounded by response divergence between the activation and apprehension/worry items, we analyzed changes in the individual items at the preexercise, during exercise, and

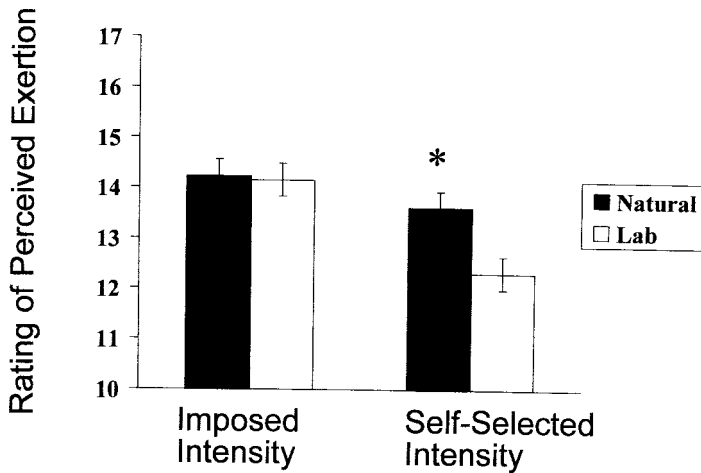


Figure 1 — Means and standard errors for ratings of perceived exertion in the naturalistic and laboratory environments for the imposed-intensity and self-selected intensity groups. * $p < .01$

Table 2 SAI and AS Responses Before, During, and Following the Three Conditions

Variable	Naturalistic			Laboratory			Quiet Rest		
	M	(SD)	ES	M	(SD)	ES	M	(SD)	ES
State Anxiety									
Pre	35.2	(7.1)	—	35.1	(7.4)	—	36.2	(8.4)	—
During	39.2	(7.4)*	.55	35.3	(6.8)	.03	33.6	(8.1)	-.31
P-5	30.9	(7.1)*	-.61	31.2	(6.1)*	-.58	32.9	(9.4)	-.37
P-60	32.7	(9.4)	-.30	29.9	(6.6)*	-.74	34.7	(9.6)	-.17
P-120	33.1	(9.7)	-.25	29.8	(7.2)*	-.73	33.9	(9.9)	-.25
P-180	34.8	(8.9)	-.04	32.5	(8.2)	-.33	33.6	(9.7)	-.29
Activation									
Pre	4.1	(1.7)	—	3.8	(1.8)	—	4.2	(1.3)	—
Dur-5	5.8	(1.3)*	1.13	5.4	(1.5)*	.97	4.2	(1.5)**	.00
Dur-10	6.2	(1.4)*	1.31	6.1	(1.3)*	1.52	4.0	(1.7)**	-.13
Dur-15	6.6	(1.3)*	1.56	6.6	(1.3)*	1.85	3.7	(1.8)**	-.32
Dur-20	6.9	(1.4)*	1.75	6.7	(1.4)*	1.87	3.8	(1.8)**	-.26
P-5	5.6	(1.7)*	.88	5.4	(1.8)*	.89	3.6	(1.9)**	.37

*Significantly different from baseline value; **Significantly different from exercise.

5-min postexercise assessments. To provide a direct comparison with previous findings (Ekkekakis et al., 1999), the 10 items that compose the short form of the SAI were analyzed using a repeated-measures MANOVA. Results revealed a significant multivariate effect for Time, Wilks' $\lambda = .35$, $F(20, 67) = 6.27$, $p < .001$, as well as a significant Condition $\times$ Time interaction, Wilks' $\lambda = .35$, $F(40, 136) = 2.08$, $p < .005$. Follow-up univariate analyses demonstrated that the items calm, tense, frightened, nervous, relaxed, and jittery were responsible for the significant multivariate interaction effect.

A close inspection of the descriptive statistics summarized in Table 3 reveals that, whereas both apprehension/worry and activation items increased during exercise in the naturalistic environment, a divergent item response pattern emerged in the laboratory environment. That is, similar to the responses documented by Ekkekakis et al. (1999), activation items increased while apprehension/worry items decreased during the in-task assessment in the laboratory condition. Additionally, consistent with the previous findings (Ekkekakis et al., 1999; Katula et al., 1999), the decrease in total SAI score evident at the 5-min postexercise assessment following laboratory and naturalistic exercise was composed of a unitary reduction in both activation and apprehension/worry items. Descriptive statistics and effects sizes accompanying changes in the 10-item SAI are listed in Table 3 and changes in all 20 items of the SAI are summarized in Table 4.

Perceived Arousal

Analyses of the AS yielded significant main effects for Time, $F(5, 140) = 26.16$, $p < .001$, and Condition, $F(2, 56) = 36.76$, $p < .001$, as well as a significant Condition $\times$ Time interaction, $F(10, 280) = 15.97$, $p < .001$. Post-hoc analyses revealed that perceived arousal was significantly higher than baseline at all assessments during both exercise conditions, and this elevation persisted through the 5-min postexercise assessment. Additionally, perceived arousal during and after both exercise conditions was found to be significantly higher compared to the quiet rest condition (see Table 2).

Discussion

The primary finding of this study was that, among women with high SPA, state anxiety responses are influenced by the perceived evaluative threat present in the exercise environment. More specifically, the laboratory and naturalistic environments were associated with divergent patterns of change in state anxiety during and after exercise. Hence, whereas the anxiolytic benefits of acute exercise have been documented independent of the exercise environment (McAuley et al., 1996), the present results suggest this finding does not generalize to women who are high in SPA.

Consistent with self-presentational theory (Schlenker & Leary, 1982), an increase in state anxiety only emerged during exercise in the environment that induced the greatest perceived evaluative threat. Item-level analysis of the SAI also revealed response patterns that differed as a function of the environment in which exercise was performed. Total SAI score during the laboratory condition subsumed both increases in activation items and decreases in apprehension/worry items. Conversely, the total SAI score observed in the naturalistic environment contained increases in both apprehension/worry and activation items. Therefore,

Table 3 The 10-Item SAI Scores Before, During, and Following Acute Exercise in Each Environment

Item	Naturalistic			Laboratory		
	<i>M</i>	(<i>SD</i>)	ES	<i>M</i>	(<i>SD</i>)	ES
1. Calm						
Pre	1.93	(.74)	—	1.90	(.80)	—
During	2.63	(.72)*	.96	2.20	(.81)*	.37
Post-5	1.73	(.69)*	-.28	1.63	(.67)*	-.37
2. Tense						
Pre	1.73	(.69)	—	1.60	(.89)	—
During	2.00	(.69)*	.39	1.47	(.68)	-.17
Post-5	1.30	(.60)*	-.67	1.27	(.69)*	-.42
3. Frightened						
Pre	1.03	(.18)	—	1.27	(.69)	—
During	1.07	(.25)	.19	1.03	(.18)*	-.55
Post-5	1.03	(.18)	.00	1.03	(.18)*	-.55
4. Nervous						
Pre	1.30	(.54)	—	1.50	(.57)	—
During	1.37	(.67)	.11	1.16	(.38)*	-.71
Post-5	1.07	(.37)*	-.50	1.13	(.35)*	-.80
5. Relaxed						
Pre	2.20	(.71)	—	2.20	(.81)	—
During	2.60	(.89)*	.40	2.30	(.84)	.12
Post-5	1.77	(.63)*	-.64	1.87	(.78)*	-.42
6. Worried						
Pre	1.53	(.86)	—	1.47	(.82)	—
During	1.57	(.77)	.05	1.27	(.52)	-.30
Post-5	1.33	(.61)	-.27	1.30	(.60)	-.25
7. At Ease						
Pre	2.20	(.66)	—	2.13	(.82)	—
During	2.63	(.85)	.57	2.23	(.90)	.12
Post-5	1.86	(.68)	-.50	1.77	(.68)	-.48
8. Jittery						
Pre	1.23	(.43)	—	1.30	(.54)	—
During	1.37	(.72)	.24	1.13	(.35)*	-.38
Post-5	1.07	(.25)*	-.47	1.20	(.41)	-.21
9. Steady						
Pre	2.23	(.94)	—	2.30	(.92)	—
During	2.60	(.81)	.42	2.37	(.93)	.07
Post-5	2.00	(.74)	-.27	2.10	(.85)	-.38
10. Worrying over possible misfortunes						
Pre	1.77	(.94)	—	1.70	(.95)	—
During	1.60	(.81)	-.19	1.27	(.58)	-.56
Post-5	1.37	(.61)	-.52	1.23	(.63)	-.59

*Significantly different from baseline value.

Table 4 The 20-Item SAI Scores Before, During, and Following Acute Exercise in Each Environment

Item	Naturalistic		Laboratory	
	<i>M</i>	(<i>SD</i>)	<i>M</i>	(<i>SD</i>)
1. Calm				
Pre	1.93	(.74)	1.90	(.80)
During	2.63	(.72)	2.20	(.81)
Post-5	1.73	(.69)	1.63	(.67)
2. Secure				
Pre	1.87	(.68)	1.97	(.76)
During	2.17	(.69)	1.97	(.72)
Post-5	2.03	(.89)	2.00	(.69)
3. Tense				
Pre	1.73	(.69)	1.60	(.89)
During	2.00	(.69)	1.47	(.68)
Post-5	1.30	(.60)	1.27	(.69)
4. Strained				
Pre	1.60	(.89)	1.67	(.92)
During	2.10	(.76)	2.07	(.78)
Post-5	1.43	(.68)	1.47	(.68)
5. At Ease				
Pre	2.20	(.66)	2.13	(.82)
During	2.63	(.85)	2.23	(.90)
Post-5	1.86	(.68)	1.77	(.68)
6. Upset				
Pre	1.30	(.65)	1.13	(.35)
During	1.23	(.43)	1.10	(.40)
Post-5	1.10	(.31)	1.03	(.18)
7. Worrying over possible misfortunes				
Pre	1.77	(.94)	1.70	(.95)
During	1.60	(.81)	1.27	(.58)
Post-5	1.37	(.61)	1.23	(.63)
8. Satisfied				
Pre	2.17	(.87)	2.30	(.95)
During	2.27	(.69)	2.27	(.83)
Post-5	1.93	(.69)	1.90	(.66)
9. Frightened				
Pre	1.03	(.18)	1.27	(.69)
During	1.07	(.25)	1.03	(.18)
Post-5	1.03	(.18)	1.03	(.18)
10. Comfortable				
Pre	1.87	(.63)	1.77	(.50)
During	2.43	(.89)	2.30	(.84)
Post-5	1.87	(.82)	1.80	(.61)

(continued)

Table 4 (Continued)

Item	<u>Naturalistic</u>		<u>Laboratory</u>	
	<i>M</i>	(<i>SD</i>)	<i>M</i>	(<i>SD</i>)
11. Self-Confident				
Pre	2.30	(.92)	2.47	(.82)
During	2.33	(.71)	2.57	(.73)
Post-5	2.13	(.82)	2.40	(.72)
12. Nervous				
Pre	1.30	(.54)	1.50	(.57)
During	1.37	(.67)	1.16	(.38)
Post-5	1.07	(.37)	1.13	(.35)
13. Jittery				
Pre	1.23	(.43)	1.30	(.54)
During	1.37	(.72)	1.13	(.35)
Post-5	1.07	(.25)	1.20	(.41)
14. Indecisive				
Pre	1.67	(.76)	1.47	(.57)
During	1.43	(.63)	1.43	(.57)
Post-5	1.20	(.48)	1.37	(.56)
15. Relaxed				
Pre	2.20	(.71)	2.20	(.81)
During	2.60	(.89)	2.30	(.84)
Post-5	1.77	(.63)	1.87	(.78)
16. Content				
Pre	2.13	(.82)	2.17	(.70)
During	2.20	(.76)	2.23	(.77)
Post-5	1.83	(.79)	1.87	(.73)
17. Worried				
Pre	1.53	(.86)	1.47	(.82)
During	1.57	(.77)	1.27	(.52)
Post-5	1.33	(.61)	1.30	(.60)
18. Confused				
Pre	1.23	(.50)	1.13	(.43)
During	1.27	(.45)	1.13	(.43)
Post-5	1.13	(.43)	1.17	(.38)
19. Steady				
Pre	2.23	(.94)	2.30	(.92)
During	2.60	(.81)	2.37	(.93)
Post-5	2.00	(.74)	2.10	(.85)
20. Pleasant				
Pre	1.96	(.76)	1.83	(.65)
During	2.33	(.66)	2.13	(.82)
Post-5	1.73	(.69)	1.73	(.52)

while the divergent item response pattern may have masked an anxiolytic effect during exercise in the laboratory environment, the increase in apprehension/worry items suggests that exercise in the naturalistic environment was associated with an anxiogenic response.

In light of the recent criticisms regarding the validity of SAI, the item-by-item findings may have important implications for the use of this inventory in the context of acute exercise. The observation of opposing patterns of change in the SAI items during exercise in the laboratory environment yields support for the contention that the total SAI score can be confounded when administered during acute exercise (Ekkekakis et al., 1999; Katula et al., 1999; Rejeski et al., 1991). Nevertheless, both activation and apprehension/worry items of the SAI increased during exercise in the naturalistic environment, the setting that induced greater perceptions of evaluative threat. It should be acknowledged that it is not clear whether the elevation in activation items is related to the experience of anxiety or to the physical demands of the exercise stimulus. Thus the present findings echo the contention that we must disentangle anxiety-induced activation from exercise-induced changes in perceived arousal (Ekkekakis et al., 1999). In this regard, further inquiry is warranted which examines how contextual variables and individual differences may modify the influence of acute exercise upon changes in the cognitive and somatic aspects of anxiety.

The differential AS and SAI responses observed following exercise also merits discussion. Regardless of the environment or the performance of exercise of imposed or self-selected intensity, increases in the activation items of the SAI were accompanied by concomitant elevations in the AS during exercise. However, whereas perceived arousal remained significantly higher than baseline following the activity, the reduction in state anxiety exhibited 5 min after both exercise conditions was composed of a decrease in both apprehension/worry and activation items. Therefore, compared to baseline levels, items that reflect activation decreased at the same assessment interval during which elevations in perceived arousal, as measured by the AS, persisted.

Although the factors responsible for the divergent item level responses are not currently known, this pattern of change may reflect issues with the item-specificity of the SAI. Notably, it is unclear at this time the extent to which responses to items such as calm, jittery, tense, or at-ease are differentially influenced by the perception of cognitive and somatic cues. Consequently, while activation items of the SAI may be impacted by the physiological demands of the exercise stimulus, the present results also suggest that such descriptors may be interpreted as having emotional content which can vary independently of indices of perceived arousal (Watson & Tellegen, 1999).

Although perceived arousal was higher than the baseline value 5 min after both exercise conditions, elevations in perceived activation have not persisted as long following the cessation of activity in prior studies (Ekkekakis et al., 1999; VanLanduyt et al., 2000). There may be a number of plausible explanations, but it seems likely that the use of different perceived arousal measures contributed to the divergent responses. Another potential explanation involves the activity history of the samples. The participants in this study were relatively inactive, participating in two or less bouts of exercise during a typical week. By contrast, the samples in the other studies were considerably more active (Ekkekakis et al., 1999; VanLanduyt et al., 2000). Thus it is possible that the low activity level of our participants re-

sulted in elevations in perceived arousal that persisted longer following activity. Based on the differential responses observed in these investigations, the convergent validity of various measures of perceived arousal and individual differences which may influence self-reported arousal responses to exercise should be explored further.

Interestingly, although reductions in state anxiety were observed 5 min following both exercise conditions, the anxiolytic effect only persisted after exercise in the laboratory environment. The factors that contributed to the differential duration of postexercise anxiolysis observed following the naturalistic and laboratory conditions are not yet clear. However, given that in-task subjective states are related to postexercise psychological responses (Rejeski, Gauvin, Hobson, & Norris, 1995), the changes in state anxiety following exercise must be considered in the context of the responses that emerged in each environment. Recall that, compared to the laboratory environment, exercise in the naturalistic environment was associated with an increase in state anxiety and higher perceptions of evaluative threat.

These divergent in-task responses imply differential appraisals of the exercise conditions and it is possible that the transient anxiolytic response following the naturalistic condition reflects relief at leaving an environment that induced higher perceptions of evaluative threat. However, because the mechanisms responsible for the divergent postexercise anxiety responses cannot be determined from these findings, we must acknowledge that this is a limitation of the present study. That is, the exercise environments in this study have differences beyond the prospect of evaluative threat (lighting, presence of music and TV, noise, presence of windows, etc.). Therefore, while the exercise environments were manipulated to create differences in perceived evaluative threat, it is possible that variations in the environment unrelated to self-presentational concerns also contributed to the present findings; the veracity of this contention warrants further inquiry.

Consistent with previous research (Dishman et al., 1994), imposed-intensity exercise was significantly higher than that observed during self-selected intensity activity. Accordingly, the present findings support the contention that employing imposed intensities that are chosen arbitrarily by researchers may not accurately reflect the exertion that individuals would choose during their own exercise participation (Dishman, 1994; Gauvin et al., 1996; Morgan, 1997). Nevertheless, state anxiety responses to exercise were not influenced by the performance of self-selected or imposed-intensity exercise.

When considering this finding in the context of the telic/paratelic metamotivational states described in reversal theory (Apter, 1989), one explanation may be that, for women with high SPA, qualities of the environment that elicit perceptions of self-presentational threat are more influential than the characteristics of the exercise stimulus itself in determining the state anxiety responses during exercise. For example, obese women reported anxiety associated with the prospect of being negatively evaluated by others as more aversive than the exercise-induced exertion (Bain et al., 1989). Thus it is possible that self-presentational concerns among women high in SPA induce a telic metamotivational state (i.e., focusing on the goal of impression management) during exercise. Once in the telic state, the apprehension of high SPA individuals arising from the fear of negative bodily evaluation would be expected to become more salient than the sensations emanating from the exercise stimulus itself in determining state anxiety responses to exercise.

However, other explanations warrant consideration. It has been contended that the exteroceptive and interoceptive contributions to the interpretation of the exercise stimulus vary with the metabolic demands of the bout (Ekkekakis & Petruzzello, 1999; Rejeski, 1985). Thus, whereas subjective appraisal may take primacy during less intense exercise, physiological symptoms may become more salient as intensity approaches maximum. Because the average exercise intensity in this study for both the self-selected and imposed-intensity groups was relatively moderate (63% to 78% of age-predicted HR max), it is possible that the level of exertion may not have been high enough to shift the participants' focus from qualities of the environment that induce self-presentational threat to the physiological cues generated by the exercise stimulus. It is also possible that any potential influence of the modest difference in intensity between the two groups could have been superceded by the fact that the self-selected intensity group was permitted to choose their exercise intensity while the imposed-intensity group was not. Although each of these explanations remains tenable, delineating the influence of intensity preferences upon the affective responses to exercise has important public health implications and warrants further inquiry.

In addition to the state anxiety responses, perceived exertion was also found to be influenced by the exercise environment. Among women randomized to the self-selected intensity group, RPE was significantly higher during exercise in the naturalistic environment. Consistent with this finding, there was also a trend toward intensity being higher during self-selected intensity exercise undertaken in the naturalistic environment. It is possible that the self-presentational qualities of the environment influenced the exertion at which the self-selected group chose to exercise. For example, Baumeister (1982) proposed that maintaining consistency of one's public self with one's desired self is a primary motive of self-presentation. Furthermore, Leary and Kowalski (1990) suggested that tailoring one's public image to the perceived preferences of others and attempting to project an image consistent with the demands of the situation are primary sources of information used to create an impression.

When considering the RPE responses in this study within the framework of self-presentational theory, it is possible that the higher RPE observed during self-selected intensity exercise in the naturalistic environment represents an attempt to conform to the level of exertion perceived to be the norm for the other exercisers in the facility. Although the observed faster running speeds in the presence of perceived social evaluation supports this assertion (Worringham & Messick, 1983), more research is needed to elucidate the influence of perceived self-presentational threat on the selection of intensity and sense of effort during acute exercise.

Given that the exercise-induced state anxiety responses differed as a function of the perceived evaluative threat in the environment, the current findings have implications for the promotion of physical activity. Many individuals engage in acute exercise as a way to regulate mood (Thayer, Newman, & McClain, 1994), and subjective responses during activity have been found to be predictive of postexercise changes in psychological states (Rejeski et al., 1995). Consequently, the present findings suggest that exercising in a setting where it is possible to minimize aspects of the environment that foster perceptions of evaluative threat may contribute to the affective benefits of exercise among women with high SPA.

There is also evidence that the psychological distress which accompanies

the prospect of evaluative threat can deter exercise participation (Bain et al., 1989; McAuley, Bane, Rudolph, & Lox, 1995). Therefore, in regard to the adoption and adherence to exercise, the present findings suggest it may be better for less active women with high SPA to exercise in environments where the prospect of evaluative threat can be modified. In support of this contention, when compared to group-based exercise therapy, superior exercise participation was observed for home-based exercise among a sample of middle-aged obese women (Perri, Martin, Leermakers, Sears, & Notelovitz, 1997). Although Perri and colleagues concluded it may have been the inconvenience of the group-based program that was responsible for the differences in exercise adherence, it is possible that greater control over a social environment that induces perceptions of evaluative threat (i.e., option of exercising in more private settings) may also have been a factor.

The present findings also suggest that further exploration of the exercise/anxiety relationship within the framework of self-presentational theory is warranted. Notably, it is possible that differences in self-presentational processes such as impression motivation and impression construction may influence the anxiolytic benefits of acute exercise. Although the effect that such individual differences may have on nomothetic conceptualizations of the exercise/anxiety relationship remains unclear (VanLanduyt et al., 2000), recent advances in the measurement of such impression-management tendencies should help elucidate the role of self-presentational processes in the psychological consequences of exercise behavior (Conroy et al., 2000).

In conclusion, the present findings suggest that for women high in SPA, state anxiety responses to acute exercise differ with the perceived evaluative threat in the exercise environment. Therefore, whereas the environment has not been found to influence the anxiolytic effect of exercise previously (McAuley et al., 1996), these findings may not generalize to women with high SPA. The present findings and the emerging consensus of the role of self-presentational concerns in exercise behavior reveals that the relationship between SPA, exercise, and psychological well-being remains a viable area for further inquiry.

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